

CPS I — Exercise Sheet 6

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1	2	3	Σ
6/7	2.5/5	3.5/5	12.5/17

Exercise 1: Executions, Paths and Traces 6/7

2/2 • (a)

- $s_1 \xrightarrow{\alpha} s_1$ ✓
- $s_0 \xrightarrow{\alpha} s_1$ ✓
- $s_1 \xrightarrow{\alpha} s_1 \xrightarrow{\alpha} s_1 \dots$ (infinite) ✓
- $s_0 \xrightarrow{\alpha} s_1 \xrightarrow{\alpha} s_1 \dots$ (infinite) ✓

$$= (s_1 \xrightarrow{\alpha} s_1)^\omega$$

$$= s_0 \xrightarrow{\alpha} (s_1 \xrightarrow{\alpha} s_1)^\omega$$

1/1 • (b)

- $s_0 \xrightarrow{\alpha} s_1$: Only 1 execution.
- $s_0 \xrightarrow{\beta} s_2$: At each s_2 , choose β (stay) or γ (go to s_3).

Infinitely many executions. ✓

0.5/1 • (c)

Infinitely many

explanation missing: ~0.5

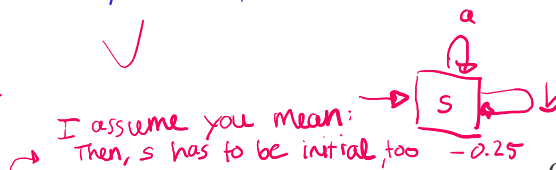
0.75/1 • (d)

- Trace 1: $\emptyset, \{a\}, \{a\}, \{a\}, \dots = \emptyset \{a\}^\omega$
- Trace 2: $\emptyset, \{b\}, \{b\}, \{b\}, \dots = \emptyset \{b\}^\omega$

2 traces. ✓

1.75/2 • (e) Bonus

Yes, it is possible. ✓



Consider a single state s with two distinct self-loops on s : $s \xrightarrow{a} s$ and $s \xrightarrow{b} s$ (where $a \neq b$).

- **Only 1 path**: A path is an infinite sequence of states. Since there's only one state s , the only possible state sequence is $s \rightarrow s \rightarrow s \rightarrow \dots$ ✓
- **Infinitely many executions**: At each step, we can choose either transition a or b , leading to infinitely many distinct sequences ✓

Exercise 2: Linear-Time Properties 2.5/5

• (a)

1.5/2.5

- 0.5*
- $T_1 \{A_0 A_1 A_2 \dots \mid A_0 = \{a\} \wedge A_1 = \{b\}\}$ ✓ *should be the other way: $\{a\}\{b\}\{a\}\{b\}\dots$*
 - $T_2 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. (i \text{ odd} \implies a \in A_i) \wedge (i \text{ even} \implies b \in A_i)\}$ *(or define trace as: $A_1 A_2 A_3 \dots$)*
 - $T_3 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. (A_i = \emptyset \implies (a \in A_{i+1} \vee b \in A_{i+1}))\}$ ✓
 - $T_4 \{A_0 A_1 A_2 \dots \mid A_0 = \{b\}\}$ ✓ *propositional logic operator used on sets*
 - $T_5 \{A_0 A_1 A_2 \dots \mid A_0 = \{a\} \wedge A_1 = \{a\}\} \wedge \{\exists k. \forall i > k. a \notin A_i\}$ *set*
- syntax: -0.5*
- (b)
 - $T_1 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$ ✓
 - $T_2 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$ *was used already ... -0.5 x 3*
 - $T_3 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$
 - $T_4 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$
 - $T_5 \{A_0 A_1 A_2 \dots \mid \exists i \in \mathbb{N}. a \in A_i \wedge b \in A_i\}$ ✓
- 1/2.5*

Exercise 3: Starvation Freedom *3.5/5*

- 0.5/2*
- (a) $\text{LIVE}' \subseteq \text{LIVE}$
If a trace $\pi \in \text{LIVE}'$, then for every $i \in \mathbb{N}$:

if $\text{wait1} \in A_i$ then $\exists j \geq i \wedge \text{crit1} \in A_j$

and similarly for process 2. *what if it doesn't?*

If **wait1 occurs infinitely often** ($\exists^\infty i. \text{wait1} \in A_i$), then **infinitely many corresponding j exist** *why do they have to be distinct?*
such that $\text{crit1} \in A_j$.

Hence:

$\pi \in \text{LIVE}$

- (b)
Consider the trace:

1/1

$\pi = \{\text{wait1}\}, \emptyset, \emptyset, \dots$ ✓

- **LIVE:**
wait1 occurs only once, so the antecedent of LIVE is false $\rightarrow$ implication holds $\rightarrow$

$\pi \in \text{LIVE}$

- **LIVE':**
At $i = 0$, $\text{wait1} \in A_0$ but there is no $j \geq 0$ with $\text{crit1} \in A_j \rightarrow$

$\pi \notin \text{LIVE}'$ ✓

1/1 • (c) False

In mutual exclusion algorithms with starvation freedom (semaphore or Peterson), every wait is eventually followed by crit.

This means LIVE' automatically holds

1/1 • (d) False

The same reason in (a)

